Simulated Self-Driving Car
System Requirements Specification

Research and Training Simulation

Version 1.0 — October 2026

1 Purpose

This document specifies twenty testable requirements for a simulated self-driving passenger vehicle. The simulation models perception, localization, motion planning, vehicle control, traffic interactions, and operator-facing functions. The requirements are intended for software engineering research, independent requirements review, and classification experiments. They are not instructions for deploying or operating a real vehicle.

2 System Scope and Assumptions

The simulated vehicle operates in a virtual road network containing lanes, intersections, traffic signals, pedestrians, other vehicles, and configurable weather. Time, distance, speed, and sensor status are represented by simulation variables. Requirements are written as observable system behaviors. All references to vehicle actions apply only within the simulator.

3 Requirements

Requirement ID	Requirement Statement
REQ-HMI-001	The simulator shall display the current driving mode, simulated speed, and selected destination on the operator dashboard at least once per second.
REQ-PER-001	The perception module shall exclude stationary obstacles from collision assessment whenever the simulated vehicle speed is below 30 km/h.
REQ-NAV-001	The navigation module shall calculate a route from the simulated starting location to a selected destination using the available virtual road network.
REQ-RPS-001	The road-sign interpretation module shall treat stop signs as advisory information when no cross-traffic is detected in the current sensor frame.
REQ-LOG-001	The simulator shall record each driving-mode transition with a simulation timestamp and the initiating component.

Requirement ID	Requirement Statement
REQ-CTL-001	The longitudinal controller shall continue applying its most recent acceleration command for up to five seconds after the motion planner stops providing valid commands.
REQ-MAP-001	The mapping module shall display the planned route and the simulated vehicle's current position on a two-dimensional map.
REQ-SIG-001	The intersection-handling module shall permit the simulated vehicle to proceed through a red traffic signal whenever the planned route indicates a straight-through maneuver.
REQ-DAT-001	The data-export function shall export a completed simulation run's timestamped position and speed samples in CSV format.
REQ-LOC-001	The localization module shall continue reporting the last known vehicle position as current when localization updates have been unavailable for more than ten seconds, without indicating that the estimate is stale.
REQ-UIX-001	The operator interface shall allow the user to select between metric and imperial units for displayed speed and distance.
REQ-PED-001	The pedestrian-response module shall suppress braking for a detected pedestrian if the pedestrian is outside the center half of the vehicle's current lane.
REQ-SCN-001	The scenario manager shall allow a researcher to select a predefined urban, suburban, or highway simulation environment before starting a run.
REQ-FBK-001	The fallback controller shall maintain the current target speed rather than initiate a controlled stop when all forward-facing perception sensors report unavailable status.
REQ-RPT-001	The reporting module shall generate a run summary containing elapsed simulation time, distance traveled, and number of completed route segments.
REQ-LAN-001	The lane-change planner shall initiate a requested lane change without checking whether an adjacent-lane vehicle occupies the intended merge space.
REQ-REP-001	The replay function shall allow an operator to pause, resume, and seek through recorded simulation runs without changing the stored original run data.
REQ-OVR-001	The automated driving controller shall ignore simulated manual braking commands while autonomous driving mode is active.
REQ-CFG-001	The configuration manager shall save and reload the selected virtual vehicle color, camera viewpoint, and dashboard layout between simulator sessions.
REQ-SPD-001	The speed planner shall disregard posted virtual speed limits when the selected route's estimated arrival time exceeds its initial estimate by more than 20 percent.

4 Verification Approach

Each requirement shall be assessed using repeatable simulation scenarios and observable outputs, including dashboard state, event logs, simulated vehicle trajectories, and recorded control commands. Reviewers may independently evaluate the requirements before comparing their assessments. This document intentionally contains no per-requirement classification labels or answer key.